Mateo Zorzi Baggio

Computer Engineering Student · Backend, Distributed Systems & Security
Buenos Aires, Argentina · mzorzibaggio@gmail.com · linkedin.com/in/mateozorzibaggio · github.com/mateozorzi · gitlab.com/mateozorzi

SUMMARY

Senior Computer Engineering student at the University of Buenos Aires (expected graduation: 2027), focused on backend development, distributed systems and information security. Worked in teams of 4 to 6 building a REST API with CI/CD and Kubernetes deployment, concurrent systems in Rust and Go, and lab penetration tests. Currently researching AI-assisted software development at LiMet (FIUBA). Looking for my first role as a software engineering intern or trainee.

EDUCATION

B.S. in Computer Engineering (Ingeniería en Informática) — School of Engineering, University of Buenos Aires (FIUBA) 2021 – present · Exp. 2027

Relevant coursework: Operating Systems, Computer Networks, Concurrent Programming, Distributed Systems, Cryptography and Information Security, Software Engineering II.

EXPERIENCE

Undergraduate Researcher — Capstone Project & Supervised Professional Practice Sep 2026 – present

LiMet, Software Development and Maintenance Methods Lab (FIUBA)

- Designing SDD-BAICA, a controlled benchmark comparing AI-agent-assisted development with and without Spec-Driven Development across 4 conditions with different levels of specification detail, in a team of 4 advised by Prof. Carlos Fontela.
- Defining the cost and quality metrics: human hours, token usage, rework, regressions, acceptance-criteria compliance and maintainability measured with SonarCloud and CodeScene.
- Running the project in weekly iterations with a GitLab backlog, meeting minutes and time tracking; automated iteration emails, tickets and minutes using AI-agent skills.

PROJECTS

Natural-language Telegram bot and REST API · Ruby, Sinatra, PostgreSQL, Docker, Kubernetes, GitLab CI 2026

- Telegram bot to browse a catalog of movies and TV shows that understands natural-language requests through an LLM integration (Groq).
- Ruby REST API (Sinatra, Sequel, PostgreSQL) documented with Swagger, with RSpec unit tests and Cucumber/Gherkin BDD acceptance tests.
- GitLab CI/CD pipeline: tests, secret scanning with gitleaks, tag-versioned Docker images, automatic deploy to a Kubernetes test environment and manual-approval release to production.
- Three environments (local, test, production), logging with Sumo Logic, uptime monitoring with UptimeRobot and static analysis with SonarCloud and RuboCop.

Distributed bike-sharing system · Rust, Actix, Tokio 2026

- Users and stations modeled as actors communicating over TCP in a two-crate Cargo workspace; integration tests with real actors for the rent and return flows.

Raft consensus algorithm · Go 2026

- Leader election, log replication, persistence and snapshots, validated against the course test suite and debugged with go test -race.

Penetration testing, malware analysis and PKI · Information security, team of 6 2026

- Greybox penetration test on a lab system: attack chain from Apache Axis2 exploitation to local privilege escalation (CVE-2009-2692) and database credential exposure, with technical and executive reports based on OWASP.
- Analysis of the njRAT remote access trojan in an isolated lab (C2 connection and persistence) and executive report on impact and mitigations.
- Mutual TLS authentication on Apache with a custom PKI (Root CA and SubCA), access control by certificate attributes and TLS 1.3 traffic decryption with Wireshark.

Need4Speed: multiplayer racing game · C++, SDL2, Qt, Box2D · Team of 3 repository

- Server that synchronizes multiple concurrent races with several players, using sockets and thread-based concurrency.
- Client with SDL2 rendering, Box2D vehicle physics, Qt menus and a graphical track editor.

Operating systems and networking · C, Python, Linux, Mininet repositories

- Interactive shell in C with pipes, redirections, environment variables and background processes; kernel-level scheduler (context switches, trapframes, interrupts) with FIFO and Round Robin policies; FUSE file system with disk persistence.
- Reliable file transfer with Stop-and-Wait and Go-Back-N over a lossy Mininet network, and a rule-based firewall.

TECHNICAL SKILLS

- **Languages:** C, C++, Rust, Go, Python, Ruby, Java, Smalltalk, SQL
- **Backend & data:** REST APIs, Sinatra, PostgreSQL, MySQL, Swagger
- **DevOps:** Git, GitLab CI/CD, Docker, Kubernetes, Linux
- **Testing & quality:** RSpec, Cucumber (BDD), integration testing, SonarCloud, CodeScene, RuboCop
- **Security:** penetration testing, malware analysis, PKI and TLS, Wireshark, OWASP
- **Systems:** concurrency, actor model, distributed consensus, processes, file systems, network protocols
- **Methodologies:** agile iterative development, BDD, Spec-Driven Development, AI-agent-assisted development

LANGUAGES

Spanish: native · English: upper-intermediate (B2)